

Why Contrastive Vision Models Learn Sine-Wave Filters

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Antonio Torralba and Yair Weiss have worked out why simple vision models trained with [contrastive learning](#), which teaches a model to keep altered views of the same image close and different images apart, often discover sine-wave filters. For image collections with stationary statistics, meaning the same broad patterns recur across locations, they derive an optimal one-layer convolutional neural network. Its 256 filters measure selected spatial frequencies, a

nonlinearity transforms them, global averaging removes position, and a 32-dimensional linear layer partially whitens the result so common patterns count less. A water-filling calculation chooses which frequencies matter from the dataset's average power spectrum. Tests covered six synthetic collections plus CIFAR-10, CIFAR-100, and ImageNet. With simple crops and several other augmentations, gradient descent learned the predicted filters. More modern augmentation recipes were the notable exception, producing more localized filters.

This gives a mechanism for an old puzzle. [SimCLR showed](#) that basic crops, colour changes, and blur can build useful visual representations, while earlier work found that [training on structured noise](#) can transfer to real-image recognition. Frequency statistics may explain part of both results. The theory does not explain today's full systems, though. Most experiments used grayscale images, a shallow network, and a custom loss that matches the standard InfoNCE loss only under a Gaussian assumption. The authors also report that these models are far from state of the art. This is an explanation of a clean base case, not a complete theory of deep networks or vision transformers.

Sources:

- [A Theory of Contrastive Learning with Natural Images \(arXiv\)](#)
- [A Theory of Contrastive Learning with Natural Images \(ICML 2026\)](#)
- [SimCLR: A Simple Framework for Contrastive Learning](#)
- [Learning to See by Looking at Noise](#)

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